CSE331_Assignment 2_Fall 2025

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CSE331 Assignment 2
Fall 2025
Total Marks: 50
Due: 31/12/2025, 11:59 PM

Your solutions need to be handwritten. After writing down your solutions, scan and compile your answers into a single pdf file. Then submit in the following Google Form:
https://forms.gle/GpTnzVbMyQgnQBNN8

Question 1
[20 Points]

Use the Pumping Lemma to demonstrate that the following languages are not regular.

a) $L1 = \{w \in \{0, 1\}^*: 0^i1^j \text{ where } i \leq j\}$
b) $L2 = \{w \in \{a, b, c\}^*: a^ib^jc^{k+2} \text{ where } i = k \text{ and } i, j, k \geq 0\}$
c) $L3 = \{w1\#w2 \text{ such that } |w1| = 2.|w2| , \text{ where } w1, w2 \in \{0, 1\}^* \}$
d) $L4 = \{w \in \{a\}^*: a^{2^n} \text{ where } n \geq 0\}$
e) $L5 = \{w10^i : w \in \{0, 1\}^* , i \geq 0 \text{ and } |w| \geq i\}$

Question 2
[20 Points]

Give a context-free grammar for each of the following languages.
Consider, $\Sigma=\{0,1\}$.

A. The language of strings that start with 1
B. The language of strings of the form WW^R [W^R is the reverse of W]
C. The language of strings that contain the substring 001
D. The language $\{0^n10^n \text{ where } n \geq 0\}$
E. The language $\{0^i1^j \text{ where } i \leq j\}$
F. The language $\{1^i01^j01^{i+j} \text{ where } i, j \geq 0\}$
G. The language $\{0^{3n}w1^{2n}$ where $n \geq 0$ and w is a string that contains at least two 0's}
H. The language $\{0^i10^j \mid i \text{ is a multiple of three where } i, j \geq 0\}$
I. The language $\{0^i10^j \mid j = 2 + 3i \text{ where } i, j \geq 0\}$
J. The language $\{1^i01^j \mid j \text{ is a multiple of four or } i = 3 + 2j \text{ where } i, j \geq 0\}$

Question 3
[10 Points]

1. Suppose, you are given the following grammar. $\Sigma=\{0,1\}$

$$S \rightarrow SS \mid 0S1 \mid 1S0 \mid 01 \mid 10$$

   a) Show that the grammar is ambiguous by demonstrating two different parse trees. [Hint: String 011010]
   b) Find two strings of length six in the grammar above where exactly one parse tree is possible.

2. Suppose, you are given the following grammar. $\Sigma=\{a,b\}$

$$S \rightarrow PE$$
$$P \rightarrow aPa \mid bPb \mid a \mid b$$
$$E \rightarrow aaE \mid abE \mid baE \mid bbE \mid \varepsilon$$

   a) Give a leftmost derivation for the string abababb and sketch the parse tree corresponding to that derivation.
   b) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (a)) for the same string abababb.